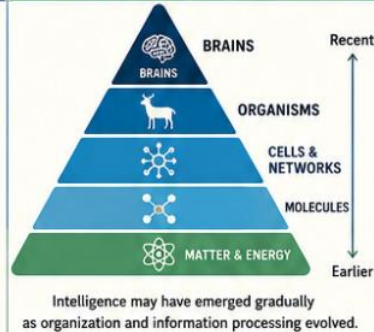


Perhaps the history of intelligence began billions of years before the first nervous system.

COMPUTATION $\neq$ INTELLIGENCE

Many biological systems
show intelligent
behaviour without brains.



Intelligence = Using information to select adaptive actions under uncertainty.

Key traits: Coordination, Division of Labour, Collective Intelligence

Emergent capacity to solve problems under uncertainty.

- Expands our definition of intelligence beyond brains.
- Helps us understand life at all scales.
- Offers new perspectives for agriculture, ecology and medicine.
- Inspires new approaches in AI and synthetic biology by learning from distributed biological intelligence.
- Reminds us that we are part of a much older story of organized matter becoming adaptive and intelligent.

Article 7 of 12
Rethinking Biology: Essays on
Information, Intelligence and Life

Perhaps the history of intelligence began billions of years before the first nervous system.

There is a question that sounds almost absurd at first:

When did intelligence begin?

The conventional answer would probably involve the evolution of nervous systems.

Perhaps intelligence began with simple neural circuits. Perhaps it became more sophisticated with brains. Perhaps it reached its highest expression with the human cerebral cortex.

This seems reasonable because, in our everyday experience, intelligence is associated with brains.

We think.

We remember.

We compare alternatives.

We anticipate consequences.

We make decisions.

We solve problems.

And all of these activities appear to depend upon the brain.

But there is a problem with this picture.

Long before there were brains, life was already doing many things that we now associate with intelligence.

A bacterium encounters an unfavorable chemical environment and changes its behaviour.

A cell detects damage and activates repair mechanisms.

An immune system distinguishes between different molecular patterns and mounts different responses.

A plant changes its growth according to light, water, temperature and neighbouring organisms.

A developing embryo responds to positional information and progressively organizes itself into an extraordinarily complex structure.

None of these systems possesses a brain.

Yet all of them face a fundamental problem:

How should a biological system respond when the future is uncertain?

Perhaps this is where our definition of intelligence needs to begin.

Intelligence is not the same as computation

In the previous articles of this series, we explored the idea that biological systems can be understood in computational terms.

Cells receive information.

They integrate signals.

They alter internal states.

They generate responses.

Molecules can participate in information processing.

And biology can be compared with artificial intelligence—not because a cell is an artificial neural network, but because both biological and artificial systems transform information into useful action.

But there is an important distinction.

Computation is not necessarily intelligence.

A calculator computes.

A thermostat processes information.

A traffic light responds to inputs.

A computer can execute an algorithm perfectly without understanding why the computation is being performed.

Intelligence seems to require something more.

It involves the ability of a system to deal with alternatives.

It must somehow distinguish between states of the world.

It must respond differently under different circumstances.

It must preserve what is useful and suppress what is harmful.

And, in many biological systems, it must do this despite incomplete information.

This suggests a broader working definition:

Intelligence may be the capacity of an organized system to use information to select adaptive actions under uncertainty.

Notice what this definition does.

It does not mention neurons.

It does not mention language.

It does not mention consciousness.

It does not even require a centralized processor.

If this definition is useful, then intelligence could be much older than brains.

The first intelligent system may have been a cell

Consider the earliest living systems.

They existed in environments that were constantly changing.

Nutrients fluctuated.

Temperature changed.

Chemical gradients appeared and disappeared.

Potentially harmful molecules were encountered.

Energy availability varied.

A primitive living system that responded identically to every environmental condition would have had little chance of surviving.

Selection would favour systems capable of distinguishing states of the environment and responding differently to them.

This distinction is crucial.

A rock can be heated.

A crystal can grow.

A chemical reaction can occur.

But a living system can **change its future behaviour according to its previous and present conditions**.

That is something fundamentally different.

The important transition may therefore not have been:

matter → brain

but something much earlier:

matter → adaptive organization.

Once adaptive organization appeared, evolution had something to work with.

From reaction to decision

We should be careful here.

Not every response is intelligent.

A molecule reacting chemically to another molecule does not necessarily constitute intelligence.

A crystal forming in a supersaturated solution is not necessarily making a decision.

So where might the distinction lie?

Perhaps in **context**.

Imagine that the same environmental signal produces different outcomes depending upon the internal state of the organism.

A nutrient may stimulate growth when energy is abundant but trigger conservation mechanisms when energy is scarce.

A signal associated with damage may produce repair in one context and programmed cell death in another.

The signal itself has not changed.

The biological interpretation has.

This is important because it means that information in biology is not simply contained in a signal.

Information depends upon the relationship between signal and system.

The same molecule can mean different things to the same cell under different circumstances.

The cell therefore cannot be understood merely as a receiver of information.

It is also an **interpreter of information**.

This idea takes us beyond the simple input-output picture of biological computation.

The cell does not need a central decision-maker

One of the most fascinating properties of biological intelligence is that it does not necessarily have a headquarters.

There is no tiny executive sitting inside the cell deciding what the cell should do.

Instead, thousands of local interactions occur simultaneously.

Molecules interact.

Regulatory networks change state.

Feedback loops amplify some signals and suppress others.

Metabolic pathways compete and cooperate.

Membranes establish compartments.

Physical constraints limit possible states.

The overall response emerges from this distributed activity.

This creates a remarkable possibility:

Intelligence may be an emergent property of interactions rather than a property of a particular component.

This idea has an important precedent in work on biological networks.

A network composed of individually rational or locally responding components can generate collective behaviour that is not obvious from any single component. Earlier work on biological networks even

examined situations in which locally consistent molecular preferences can produce collective logical paradoxes.

The lesson is broader than the particular mathematical example.

The intelligence of the system may reside in the organization of relationships.

Not in the molecule.

Not in the gene.

Not in the receptor.

But in the network.

The immune system: intelligence without a brain

The immune system provides perhaps one of the most striking examples.

There is no single "immune brain".

Instead, enormous numbers of cells and molecular interactions collectively distinguish between patterns, activate responses, suppress inappropriate reactions, generate memory and modify future behaviour.

The system is distributed.

It is dynamic.

It is adaptive.

It operates under uncertainty.

And it learns from experience.

Calling this "intelligence" may initially seem metaphorical.

But perhaps the metaphor is pointing toward something scientifically important.

If intelligence means the capacity to extract useful information from the environment and use it to alter future behaviour, then the immune system displays many properties that deserve serious examination under that framework.

The brain may therefore not have invented intelligence.

It may have **centralized and dramatically amplified an older biological capability.**

Plants force us to rethink the definition

Plants are even more uncomfortable for conventional definitions of intelligence.

They have no brain.

They have no nervous system comparable to that of animals.

Yet plants continuously integrate environmental information.

Light.

Gravity.

Water.

Temperature.

Mechanical disturbance.

Chemical signals.

Pathogens.

Neighbouring organisms.

They alter growth, development, metabolism and defence accordingly.

Earlier work associated with this research programme went even further, proposing that plants may exhibit forms of intuition despite lacking the conventional machinery associated with rational thought. That proposal is speculative and does **not** constitute established evidence for plant consciousness or quantum communication.

But we do not need the speculative part to arrive at an important question.

If a plant can integrate information from multiple environmental sources and alter its behaviour in ways that improve survival and reproduction, **what exactly are we reserving the word intelligence for?**

Perhaps the mistake is assuming that intelligence must look like human intelligence.

A plant does not need to solve a crossword puzzle.

It needs to solve the problems presented by its environment.

And those problems can be extraordinarily complex.

What about a bacterium?

The bacterium presents an even more radical case.

A bacterium does not possess neurons.

It has no brain.

Yet it must continually deal with an unpredictable environment.

It may move toward favourable chemical conditions and away from harmful ones.

It can alter gene expression.

It can change metabolic strategies.

It can interact with other organisms.

It can respond differently depending upon its physiological state.

It can participate in collective behaviours.

None of this resembles human reasoning.

But perhaps that is precisely the point.

Intelligence need not resemble human reasoning.

Evolution does not necessarily produce intelligence by making organisms increasingly similar to humans.

It produces solutions to survival problems.

Sometimes the solution is a brain.

Sometimes it is a distributed network.

Sometimes it is a biochemical feedback system.

Sometimes it is collective behaviour.

Slime moulds and the intelligence of the collective

The idea becomes even more interesting when we move from individual organisms to collectives.

A collection of relatively simple units can sometimes produce behaviour that appears surprisingly sophisticated.

This is not unique to biology.

Ant colonies can collectively solve problems that no individual ant could solve.

Bird flocks coordinate movement without a central controller.

Social insects collectively allocate resources.

Cells communicate during development to produce large-scale organization.

The important question is therefore not:

"Where is the intelligent component?"

It may instead be:

"Where is the intelligence of the system?"

The answer may be:

distributed across the interactions.

This is a fundamentally different way of thinking about intelligence.

Intelligence may be a property of organization

This leads to a hypothesis that I believe deserves much more attention:

Intelligence may be less a property of individual computational units and more a property of how information-processing units are organized, connected and constrained.

This would explain why intelligence can appear at radically different scales.

At the molecular level, interacting molecules can generate coordinated biochemical behaviour.

At the cellular level, regulatory networks integrate multiple signals.

At the organismal level, tissues and organs coordinate responses.

At the ecological level, organisms interact and collectively modify their environment.

At the social level, individual organisms generate collective behaviour.

The same fundamental principle could operate at each level:

information + interaction + memory + feedback + adaptation.

The architecture changes.

The physical substrate changes.

The timescale changes.

But the underlying informational problem remains remarkably similar.

The surprising role of memory

There is another feature that may be essential to intelligence:

memory.

Memory does not necessarily mean a brain storing explicit representations.

Biological memory can exist in many physical forms.

A molecular modification can alter future behaviour.

A change in gene expression can persist after the original signal disappears.

A cellular state can influence subsequent responses.

An immune system can respond differently after previous exposure.

A plant can retain physiological consequences of earlier environmental experiences.

In each case, the system's present behaviour depends partly upon its past.

That is enormously important.

A system that responds only to its immediate environment is reactive.

A system whose current state incorporates information from its history can become **adaptive**.

And adaptation is much closer to what we intuitively mean by intelligence.

Evolution itself is an intelligence without a brain

There is an even deeper possibility.

Perhaps intelligence should not be considered only at the level of individual organisms.

Consider evolution.

Evolution has no brain.

It does not plan.

It does not foresee.

It does not consciously optimize.

And yet, over enormous periods of time, evolution has produced extraordinarily sophisticated solutions to biological problems.

Eyes.

Wings.

Immune systems.

Photosynthetic machinery.

Molecular motors.

Neural networks.

Developmental programmes.

This does not mean that evolution is conscious.

It does not mean that evolution thinks.

But it does suggest that **problem-solving can occur without foresight.**

Natural selection explores enormous spaces of biological possibilities through variation and differential survival.

In this sense, evolution may be regarded as a distributed search process occurring across generations.

The intelligence is not located in an evolutionary "mind".

It emerges from the interaction between variation, environment, reproduction and selection.

This distinction matters.

Problem-solving does not necessarily require a problem-solver.

That may be one of the deepest lessons biology can offer artificial intelligence.

Intelligence is not consciousness

At this point we must make an important distinction.

If bacteria, plants, immune systems or evolutionary processes display forms of intelligence, does that mean they are conscious?

No—not necessarily.

Intelligence and consciousness should not automatically be equated.

A system may process information without possessing subjective experience.

A system may adapt without "knowing" that it is adapting.

A system may solve a problem without representing the problem in anything resembling a human mind.

This distinction is particularly important because some of the earlier theoretical work associated with this research programme has explored much stronger hypotheses concerning consciousness, quantum computation and a universal field of consciousness. Those ideas remain hypotheses rather than established scientific conclusions.

The present argument does not require them.

We can arrive at the idea of biological intelligence **without first solving the mystery of consciousness.**

That makes the question much more scientifically approachable.

So when did intelligence begin?

Perhaps we have been asking the wrong historical question.

We often ask:

When did animals become intelligent?

Then:

When did brains become intelligent?

Then:

When did human intelligence emerge?

But perhaps the deeper evolutionary sequence is:

When did matter begin to maintain organization?

Then:

When did living systems begin to respond adaptively to information?

Then:

When did biological systems begin to incorporate history into present behaviour?

Then:

When did distributed information processing become capable of increasingly sophisticated problem-solving?

Brains may appear relatively late in this sequence.

They may represent not the beginning of intelligence, but one extraordinary evolutionary solution to an ancient problem.

The brain may be an amplifier, not the origin

This changes how we think about ourselves.

The human brain is undoubtedly an extraordinary information-processing system.

But perhaps its deepest achievement is not that it created intelligence from nothing.

Perhaps it took an ancient biological principle and amplified it.

The cell already sensed.

The organism already adapted.

Biological networks already integrated signals.

Evolution already searched.

Collectives already coordinated.

The brain brought these capabilities together into an extraordinarily powerful architecture capable of abstraction, language, imagination and deliberate reasoning.

In that sense, the history of intelligence may be much older than the history of brains.

And perhaps much broader.

A new way to think about intelligence

The conventional picture looks something like this:

matter → cells → nervous systems → brains → intelligence

But biology may eventually suggest another picture:

matter → organization → information processing → adaptation → intelligence → brains

The difference is profound.

In the first picture, intelligence is something that appears when a sufficiently complex brain evolves.

In the second, intelligence is a property that can gradually emerge from organized matter long before brains exist.

The brain then becomes one particular—and extraordinarily powerful—implementation of a much older biological principle.

The question that follows

This brings us to an intriguing possibility.

If intelligence can exist without a brain, perhaps we should stop looking for intelligence only where we expect to find something resembling ourselves.

Perhaps we should search for it in **patterns of organization**.

In cells.

In tissues.

In immune networks.

In plants.

In microbial communities.

In ecosystems.

In collectives.

And perhaps even in the physical dynamics of biological molecules.

The challenge will be to distinguish genuine intelligence from metaphors that merely make biological systems sound intelligent.

That distinction is crucial.

We should not call every adaptive process "intelligence."

We need measurable criteria.

Can the system integrate information?

Can it distinguish contexts?

Can it modify its future responses?

Does it retain information about previous states?

Can it solve problems under uncertainty?

Can collective organization generate capabilities absent from individual components?

Can these properties be experimentally quantified?

These are questions that can move the discussion from philosophy toward science.

The bigger idea

The most important lesson may therefore be surprisingly simple:

Brains may not have invented intelligence. They may have inherited it.

If that is correct, then intelligence is not a late accident in the history of life.

It is part of a much older story—the story of matter becoming organized, interacting with its environment, retaining information, adapting to uncertainty and discovering ways to persist.

The brain is perhaps the most spectacular expression of that story.

But it may not be the beginning.

It may be one chapter in a history that started billions of years earlier.

And that raises an even more provocative question.

If intelligence can exist without a brain, could it also exist without an organism?

Could intelligence emerge from interactions among many organisms?

Could ecosystems possess forms of information processing that cannot be assigned to any individual species?

Could collective intelligence become qualitatively different from individual intelligence?

And if organized matter can process information, adapt and solve problems without a brain, **where should we draw the boundary between biological intelligence and intelligence in the wider physical world?**

That is where our journey goes next.

Article 7 of 12

Rethinking Biology: Essays on Information, Intelligence and Life

The series began by asking whether biology needs a new information theory. It has now reached a deeper question: whether intelligence itself may be a property of organized matter rather than an exclusive property of brains.